

Speeding up Digitization Using a 3D Camera

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Introduction

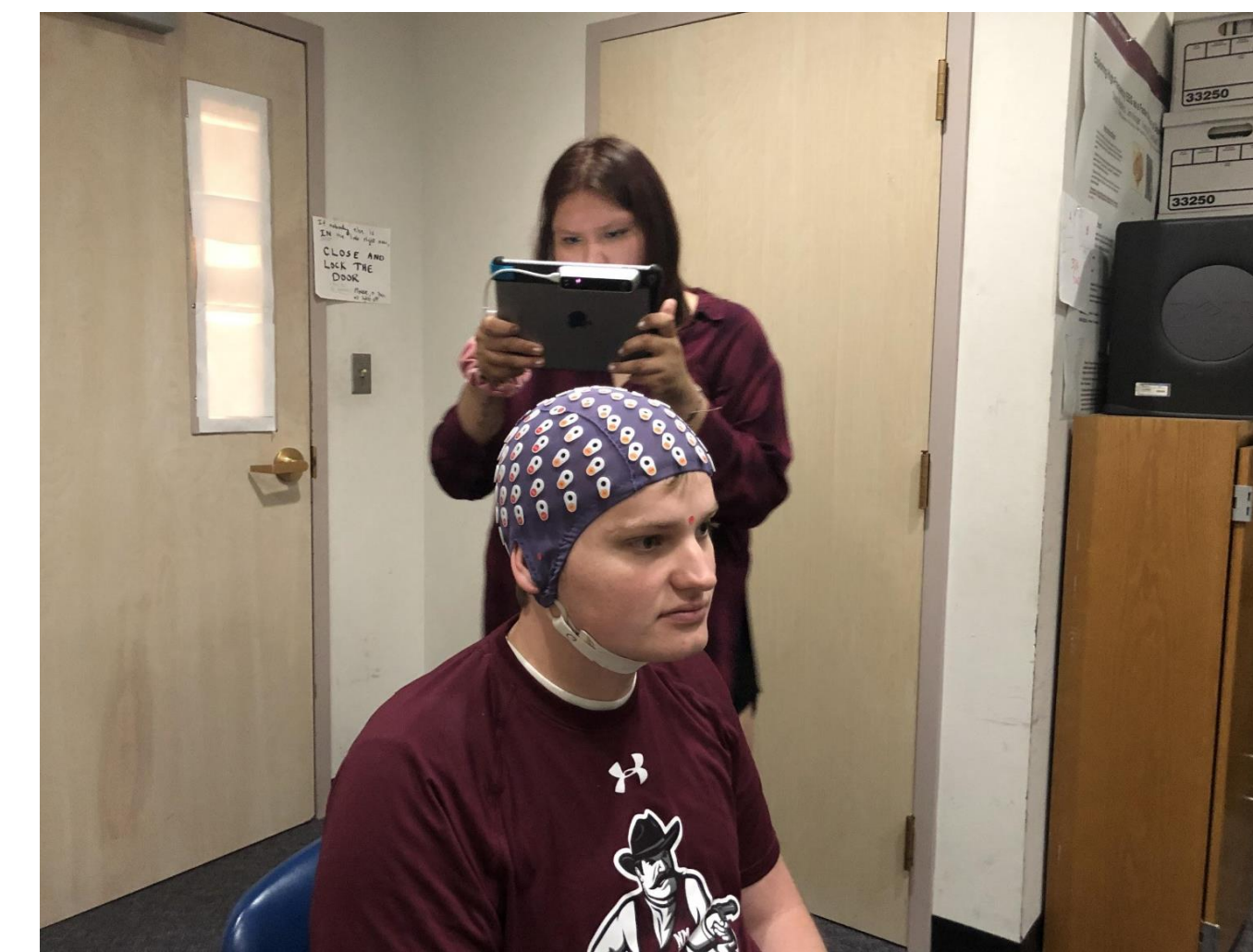
EEG Research research requires electrodes to be digitally mapped; when the electrodes are placed on the subjects head it is important to record where they are located. It is common practice to map electrodes using a Polhemus Patriot Digitizer Stylus. This method takes about 30 minutes to complete, which could lead to frustration from the subject. Recent advances in technology have introduced a new method using a 3D camera to create a model of each participant's head. This new method may reduce time to digitize electrode positions. However, testing should be done to determine the effectiveness of the 3D camera. We hypothesize that the 3D camera approach is significantly faster than the Polhemus Patriot Digitizer Stylus.

Method

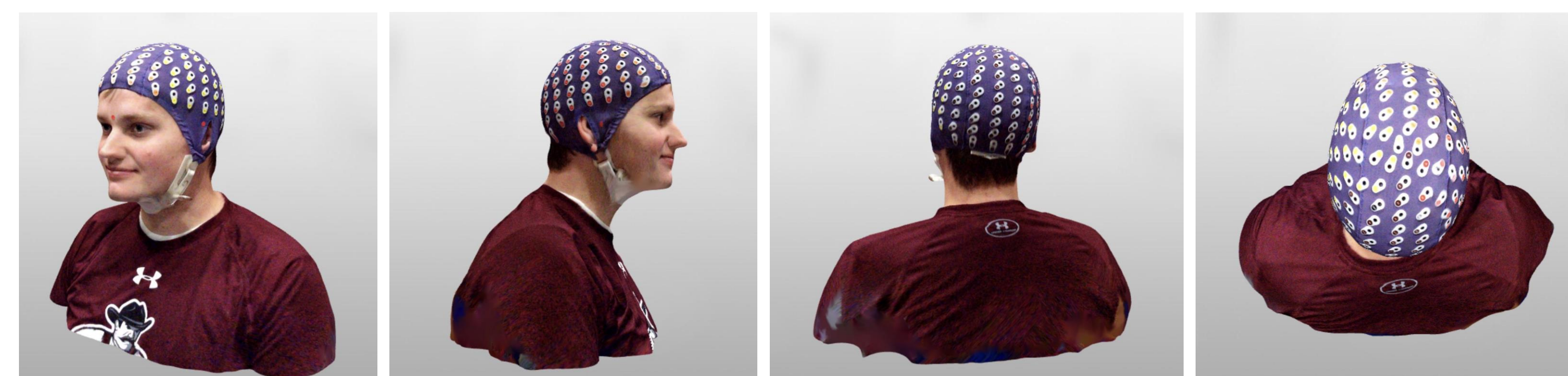
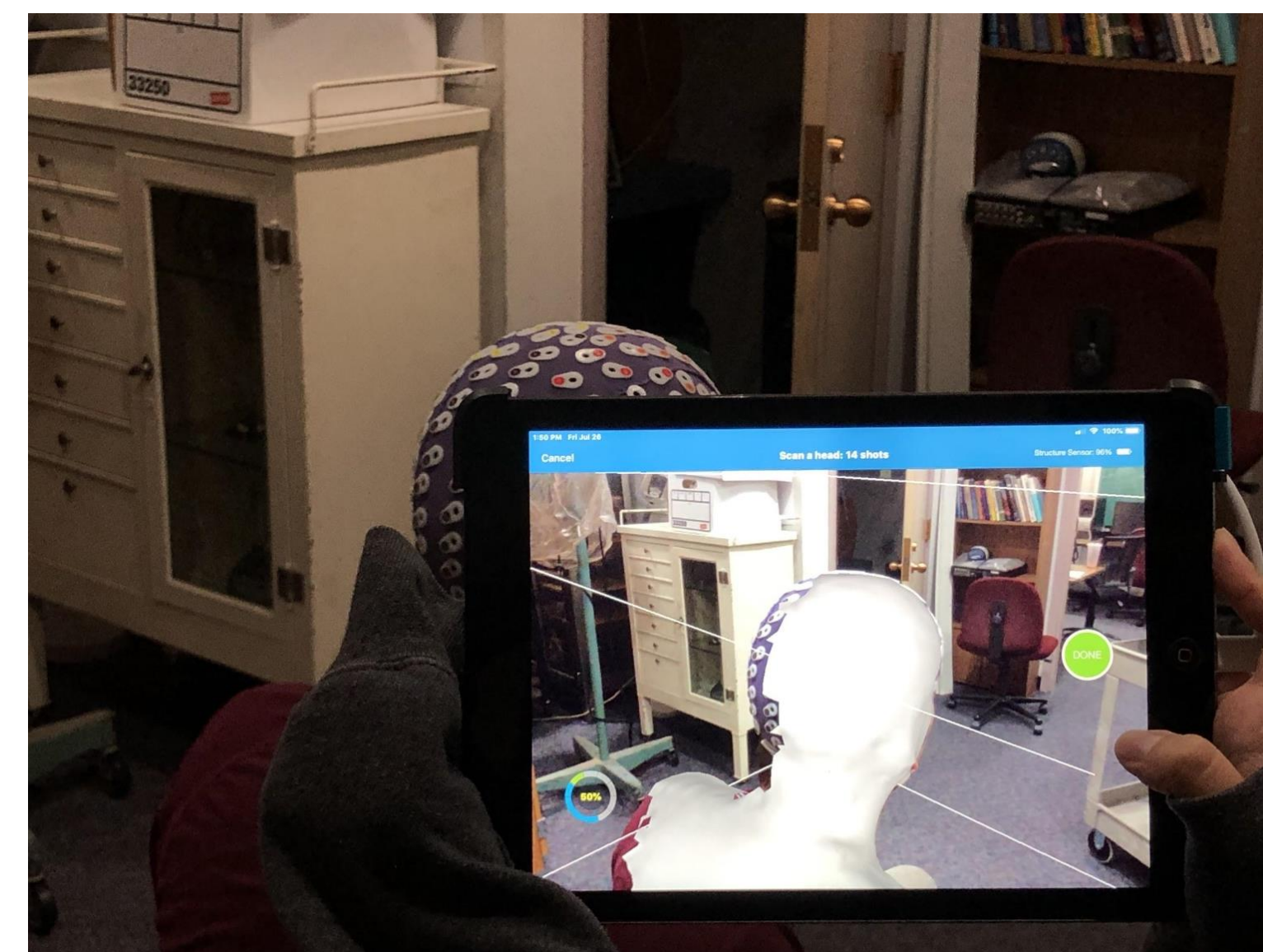
To test our hypothesis, we used a Structure Sensor connected to an iPad and Itseez3D software to capture a 3D head image which we uploaded to EEGLAB. We then used new software called *get_chanlocs* to mark the electrode positions in the head model.

Results

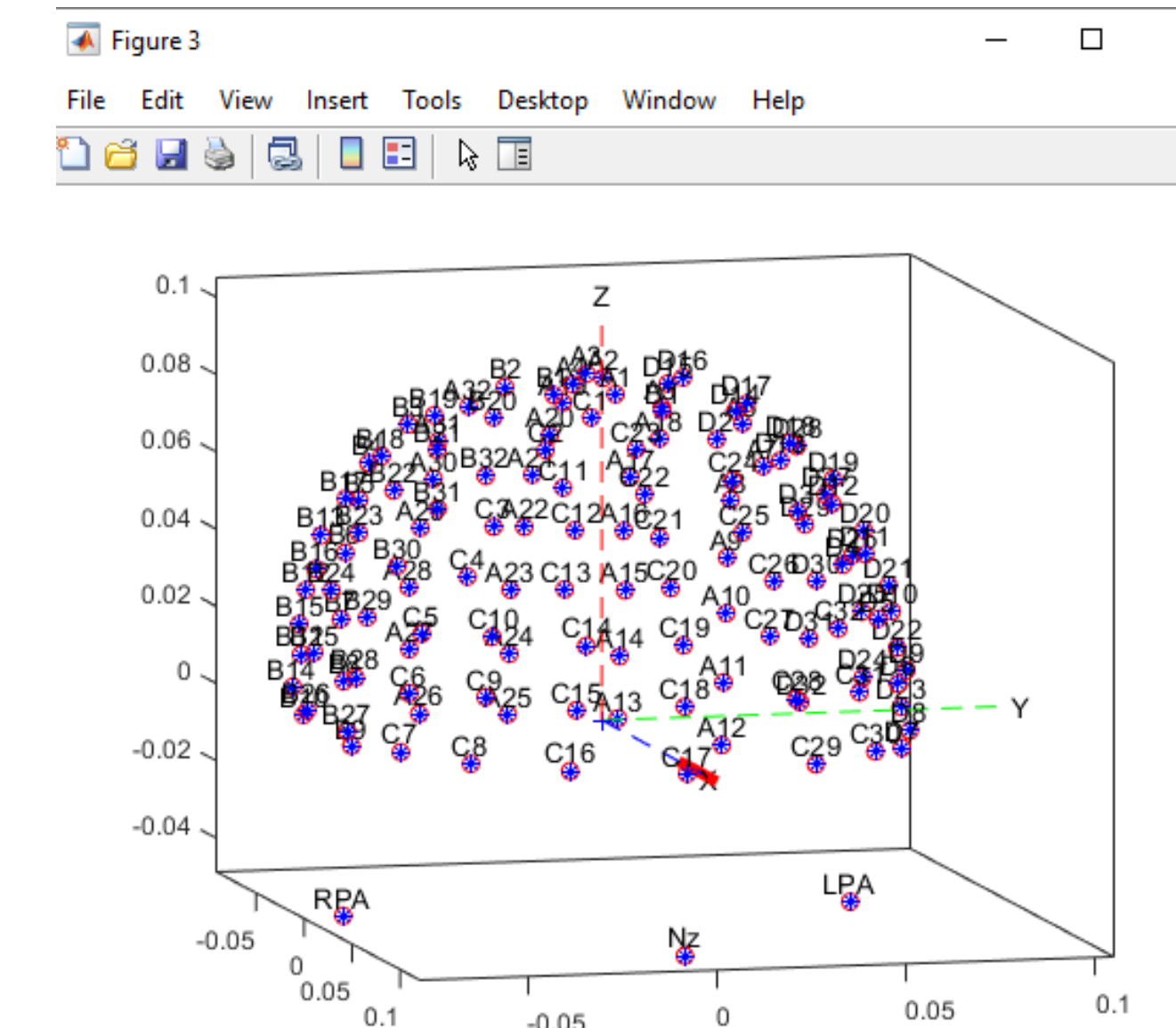
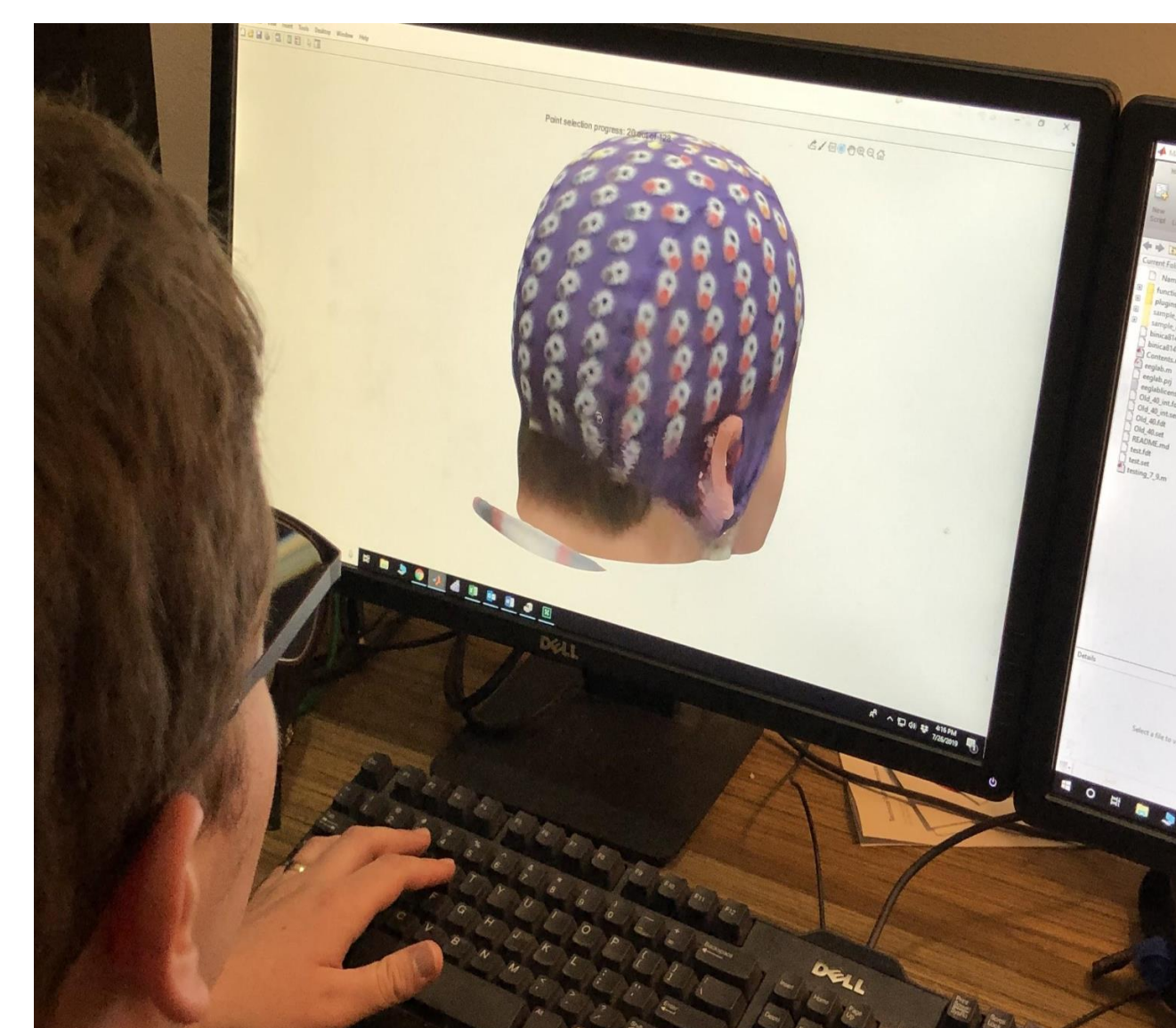
The 3D camera successfully reduced digitization time significantly. Capturing the 3D pictures of the subjects head took approximately 3 minutes. This represents a 27 minute savings in set up time.



The 3D camera approach is will be better than the Polhemus digitizer: it cuts time in half. As well as capturing a 360 image of the subjects head. This means that the subjects will get started in the experiment faster.



This new method may also provide better accuracy for electrode positions.



Discussions

- The savings in set up time with the 3D camera will provide us with much higher quality EEG data.
- Some limitations include having funds to purchase equipment, as well as time for the researchers to become oriented using the software.

References

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